

Learning Plankton Community States with Restricted Boltzmann Machines

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Abstract—Automated underwater imaging now records lake plankton every day. Turning dozens of taxon counts into a few interpretable community states is still an open problem, made harder by counts that are overdispersed, zero-heavy and partly missing. Restricted Boltzmann machines (RBMs) offer a generative latent model with exact conditionals, yet they have often been applied to ecological abundances with binary presence-absence units. We train RBMs with Negative Binomial (NB) and zero-inflated NB (ZINB) visible units on 1906 clean days of 83 taxa from Lake Greifen (2019–2024). We compare four hidden unit types over ten seeds each, stabilise training with persistent contrastive divergence, and validate the latent states against an independent archetypal analysis with five archetypes. A one-hot (Softmax) hidden layer offered up to ten states settles on about six, with either visible likelihood; its day labels match the dominant archetype (normalised mutual information 0.45, purity 71%), and three archetypes map to a single state for 94–99% of their days. An NB RBM with six Sigmoid units reproduces every archetype with a combination of its units (cosine similarity 0.94–1.00) and reconstructs a day with masked taxa better than its Softmax counterpart, whose one-hot states cost about 30% in error on the unmasked taxa. Count-likelihood RBMs, therefore, recover plankton community states that an independent method also finds, with about six states as the resolution these data support.

Index Terms—Restricted Boltzmann Machine, Negative Binomial, plankton community states, persistent contrastive divergence, imputation

I. INTRODUCTION

In order to comprehend the effects of climate change on lakes, one possible approach is to monitor the plankton biodiversity. This is due to the fact that they have rapid reproductive cycles and response time to environmental changes, making plankton abundance and diversity a good indicator of lake water quality [1]. Thus, to measure these abundances in-situ plankton monitoring technologies like the underwater Dual-magnification Scripps Plankton Camera (DSPC) imaging system offer high-frequency automated plankton monitoring in freshwater ecosystems: in particular, we consider the Lake Greifen (Switzerland) daily abundances for 83 taxa over more than five years [1]. Ecologists summarise such data with a few recurrent *community states*, or *archetypes*. A model that finds these states directly from the counts, and that can also fill in taxa missing from a day, would turn the raw stream into something a limnologist can analyze further. Generative latent-variable models learn $p(\mathbf{v})$ and, therefore, can handle both archetypal analysis and imputation. Among them, re-

stricted Boltzmann machines (RBMs) [2] have exact factorised conditionals and few parameters, and each hidden unit can be read from its weights as a feature of the data. In fact, RBMs applied to biological species can work by using binary presence-absence visible units [3] too, discarding abundances, the quantity that separates a bloom from background. In order to take into account their values, we utilise Negative Binomial (NB) likelihoods, possibly with zero inflation [4] (ZINB), which are the standard description of overdispersed, zero-rich counts. One important aspect to consider when dealing with raw abundances is to take into account the scale of the data: abundances span three orders of magnitude, and bloom and non-bloom days form well-separated modes that short Gibbs chains do not cross. We train RBMs on the Lake Greifen series (1906 clean days, 83 taxa, 2019–2024) with binary, NB and ZINB visible units and four hidden-unit types (Bernoulli, Sigmoid, Softmax, ReLU), always over 10 random seeds. We report the numerical fixes that training on raw counts requires, select the number of hidden units L on chronological and shuffled splits of the dataset and measure how many states a one-hot hidden layer actually uses. Therefore, we compare the hidden units with an archetypal analysis ($k = 5$) computed independently on the same data, and evaluate reconstruction of masked taxa and of days with structurally missing taxa. In our work we recall RBMs, Gibbs sampling and persistent contrastive divergence in Section II, the dataset, models and protocol in Section III, our findings in Section IV and, finally, the conclusions and future improvements in Section V.

II. THEORY

A. Restricted Boltzmann Machines

A restricted Boltzmann machine [2] is a Markov random field on two layers: D visible units $\mathbf{v}$, one per observed variable (here one per taxon), and L hidden units $\mathbf{h}$ that carry the latent representation. Units are coupled only across layers (Fig. 1), and a configuration has energy

$$E(\mathbf{v}, \mathbf{h}) = - \sum_i \phi_i(v_i) - \sum_j b_j h_j - \sum_{i,j} v_i W_{ij} h_j, \quad (1)$$

where ϕ_i contains the visible bias (for binary units $\phi_i(v_i) = a_i v_i$), b_j are the hidden biases and W_{ij} the couplings. The model assigns $p(\mathbf{v}, \mathbf{h}) = e^{-E(\mathbf{v}, \mathbf{h})}/Z$, with partition function $Z = \sum_{\mathbf{v}, \mathbf{h}} e^{-E(\mathbf{v}, \mathbf{h})}$ (an integral for continuous units), and the distribution of the data is the marginal $p(\mathbf{v}) = \sum_{\mathbf{h}} p(\mathbf{v}, \mathbf{h}) =$

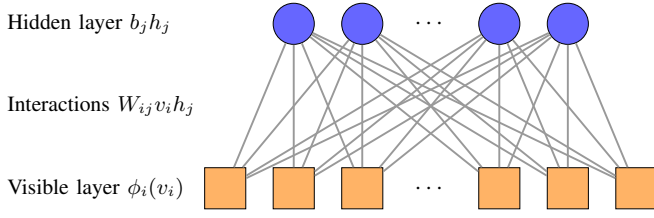


Fig. 1. An RBM is bipartite: every visible unit v_i (squares, one per taxon) is coupled to every hidden unit h_j (circles) and there are no couplings within a layer. Labels give the corresponding terms of the energy (1).

$e^{-F(\mathbf{v})}/Z$, where F is the free energy. For binary hidden units the sum over $\mathbf{h}$ factorises and

$$F(\mathbf{v}) = - \sum_i \phi_i(v_i) - \sum_j \text{softplus}\left(b_j + \sum_i W_{ij} v_i\right), \quad (2)$$

so $p(\mathbf{v})$ is known up to Z , which for $D = 83$ count variables cannot be explicitly computed.

The absence of intra-layer couplings is what makes the model usable: given one layer, the units of the other are conditionally independent,

$$p(\mathbf{h} | \mathbf{v}) = \prod_j p(h_j | \mathbf{v}), \quad p(\mathbf{v} | \mathbf{h}) = \prod_i p(v_i | \mathbf{h}), \quad (3)$$

with, for binary units, $p(h_j=1 | \mathbf{v}) = \sigma(b_j + \sum_i W_{ij} v_i)$ and $p(v_i=1 | \mathbf{h}) = \sigma(a_i + \sum_j W_{ij} h_j)$. Therefore, each hidden unit acts as a detector of one pattern of co-varying taxa, and a day is described by which detectors fire; with few hidden units each feature can be read directly from its weights.

B. Likelihood Gradient

Maximum likelihood on training days $\{\mathbf{v}^{(n)}\}$ maximises $\langle \log p(\mathbf{v}) \rangle_{\text{data}}$. For any parameter ϑ ,

$$\frac{\partial \langle \log p(\mathbf{v}) \rangle_{\text{data}}}{\partial \vartheta} = - \left\langle \frac{\partial E}{\partial \vartheta} \right\rangle_{\text{data}} + \left\langle \frac{\partial E}{\partial \vartheta} \right\rangle_{\text{model}}, \quad (4)$$

where the first average uses the data with $\mathbf{h} \sim p(\mathbf{h} | \mathbf{v})$ (positive phase) and the second the model distribution $p(\mathbf{v}, \mathbf{h})$ (negative phase). For a coupling, $\partial E / \partial W_{ij} = -v_i h_j$, so learning matches data and model correlations. The positive phase is exact thanks to Eq. (3); the negative phase is an expectation under $p(\mathbf{v}, \mathbf{h})$, as intractable as Z , and must be estimated by sampling [5].

C. Gibbs Sampling as a Markov Chain

Equation (3) gives an exact block Gibbs sampler [6]: from a state $\mathbf{v}^{(t)}$, draw $\mathbf{h}^{(t)} \sim p(\mathbf{h} | \mathbf{v}^{(t)})$ and then $\mathbf{v}^{(t+1)} \sim p(\mathbf{v} | \mathbf{h}^{(t)})$, each layer in one parallel step. The sequence $\mathbf{v}^{(0)}, \mathbf{v}^{(1)}, \dots$ is a Markov chain with transition kernel

$$T(\mathbf{v}' | \mathbf{v}) = \sum_{\mathbf{h}} p(\mathbf{v}' | \mathbf{h}) p(\mathbf{h} | \mathbf{v}). \quad (5)$$

Each half-step leaves $p(\mathbf{v}, \mathbf{h})$ invariant, so $p(\mathbf{v})$ is a stationary distribution of T ; since all conditionals are strictly positive the chain is irreducible and aperiodic, hence ergodic, and the distribution of $\mathbf{v}^{(t)}$ converges to $p(\mathbf{v})$ for any starting point. What the theory does not fix is the speed. The distance from

stationarity decays roughly as $e^{-t/\tau}$, with a mixing time τ set by the spectral gap of T , and τ grows sharply when $p(\mathbf{v})$ has well-separated modes: moving between them requires passing through low-probability configurations, which a local sampler rarely does. Plankton data are a case in point: bloom days and background days occupy distant regions of count space, and as the model learns to represent both, the energy barrier between them grows.

D. Contrastive Divergence and Persistent Chains

Running a chain to equilibrium at every parameter update is too slow, so the negative phase is approximated. Contrastive divergence (CD- k) [2] starts the chain at the current training batch and takes k Gibbs steps, usually $k = 1$. The resulting gradient is biased: if $k \ll \tau$ the samples stay near the data, and in particular a chain started from a bloom day remains in the bloom mode. The negative phase then never sees the configurations the model wrongly favours far from the data, and nothing pushes their probability down. The bias grows with the separation of the modes, which is why CD-1 becomes unreliable as L , and with it the model's ability to represent separate modes, increases (Section IV-A).

Persistent contrastive divergence (PCD) [7] keeps a set of chains alive across updates: at each step the chains continue from where they stopped, under the new parameters. This is a stochastic-approximation scheme: if the parameters change slowly compared with the mixing of the chains, the persistent chains remain close to the current model distribution, and their state accumulates many Gibbs steps over training. Therefore, they can cross between modes that a one-step chain started at the data would never reach. The approximation degrades when the learning rate is large or the mixing is extremely slow, and the training can then stay out of equilibrium in ways that shape the learned model. We use PCD with 500 epochs and a decaying learning rate (Section III-C).

E. Count Visible Units

The energy (1) admits any visible base measure ϕ_i , which yields the exponential-family harmoniums of [8]. In such a harmonium, $p(v_i | \mathbf{h})$ belongs to an exponential family whose natural parameter is the linear input $a_i + \sum_j W_{ij} h_j$. For a Negative Binomial with dispersion θ_i , the natural parameter is $\log(\mu_i / (\mu_i + \theta_i))$, which must be negative.

We instead adopt the parametrisation standard for overdispersed counts in regression: the NB mean has a log link, $\mu_i = \exp(a_i + \sum_j W_{ij} h_j)$ (Eq. (8)), so the input sets the log-mean directly and is unconstrained. The price is that this pair of conditionals, NB with log link and logistic hidden units, does not derive from a single energy of the form (1). The model is a conditional specification trained with the CD/PCD scheme of Eq. (4), in which the visible statistic v_i is replaced by the score of the NB log-likelihood with respect to the log-mean (Eq. (9)). The conditional factorisation and the block Gibbs sampler are unchanged, which is what sampling, imputation and reconstruction require. Moreover, the chain is still ergodic and has a unique stationary distribution, but

that distribution is defined by the sampler rather than by an energy, therefore the free energy (2) and pseudo-likelihoods computed from it are exact only for the binary baseline. Zero inflation [4] adds a structural-zero probability on top of the NB:

$$p(v_i | \mathbf{h}) = \pi_i \delta_0(v_i) + (1 - \pi_i) \text{NB}(v_i; \mu_i(\mathbf{h}), \theta_i) \quad (6)$$

where π controls the zero-inflation parameter while θ is the overdispersion parameter.

III. METHODS

A. Dataset and Preprocessing

We use the daily plankton abundances of the Lake Greifen imaging station [1]: 2113 days from 2019-03-21 to 2024-12-31, 83 taxa, in organisms per μL (range $[0, 0.444]$, 25.7% zeros). Total daily abundance is dominated by biology rather than instrument drift (coefficient of variation 1.29, dominant Lomb–Scargle period ≈ 365 days, stable annual median); an episode of $\approx 7\times$ the mean abundance in December 2022–February 2023 is kept as a real winter bloom, since the data descriptor reports no instrument issue in that window [1].

Four preprocessing steps are applied. (i) The 47 all-zero days are dropped. (ii) The 160 days with missing taxa are set aside as a separate test set: their gaps are not random but follow classifier-retraining blocks, in three patterns with 3, 31 and 54 missing taxa (104, 43 and 13 days respectively). (iii) Rows are not normalised, so total biomass remains a signal. (iv) Values are multiplied by $s = 1000$, which brings them to an approximately integer scale (range $[0, 444]$); at sub-unit values the gradient of the NB log-likelihood with respect to the dispersion vanishes and the dispersion never leaves its initial value. The resulting 1906 clean days are split 85/15 into training and validation in two ways. The *chronological* split validates on the last 15% of days (2024). The *shuffled* split draws validation days at random, with the run’s seed, so that every season appears in training.

We use a binary baseline to check the validity of the count models. Each taxon is binarised at its training-set median (*median*), which gives balanced units, or using a zero threshold (*zero*), depending on the model used. The zero threshold has the downside of making common taxa almost always active.

B. Models

All models share the bipartite structure of Section II-A, with $D = 83$ visible units and L hidden units, and differ in the conditional distribution of each layer.

The binary baseline has

$$p(v_i=1 | \mathbf{h}) = \sigma(a_i + \sum_j W_{ij} h_j) \quad (7)$$

The NB visible units have

$$p(v_i | \mathbf{h}) = \text{NB}(v_i; \mu_i(\mathbf{h}), \theta_i), \quad \mu_i(\mathbf{h}) = e^{\eta_i}, \quad (8)$$

with log-mean $\eta_i = a_i + \sum_j W_{ij} h_j$, mean μ_i , variance $\mu_i + \mu_i^2/\theta_i$ and a learned dispersion $\theta_i > 0$ per taxon; the log-gamma form of the NB mass function accepts the non-integer

TABLE 1. Training protocol, identical for all families unless stated.

Parameter	Value
Optimiser	RMSprop, $\beta = 0.9$, $\epsilon = 10^{-4}$
Learning rate	$0.01 \times 0.998^{\text{epoch}} (\theta, \pi: \times 0.1)$
Mini-batches	20/epoch, size $10 \rightarrow 256$ (quadratic)
Negative phase	PCD-1, 500 persistent chains
Regularisation	$\ell_1 = 10^{-4}$ on W
Epochs	500 (ZINB-Sigmoid/Softmax: 1000)
Guards	$\eta \leq 10$; ReLU $h \in [0, 5]$; $\theta \geq 10^{-4}$
Seeds	10 per (family, L , split)
Count scale	$s = 1000$

scaled counts. The goal of introducing ZINB is to try to take into account the multi-modal nature of the data, in particular for the zero (or near zero) abundances of both rare taxa and of the baseline values of many species in their inter-bloom periods. This is, also, due to the fact that the zero in abundance comes from both the camera resolution (there are species but not enough/not for enough time to trigger a measurement) and from the actual absence of the species.

Given $\mathbf{v}$, the hidden input is $x_j = b_j + \sum_i W_{ij} v_i$. We compare four types. *Bernoulli*: $p(h_j=1 | \mathbf{v}) = \sigma(x_j)$, sampled as binary. *Sigmoid*: the same Bernoulli units, trained identically, but every deterministic read-out (reconstruction, hidden representation) uses the activation probability $\sigma(x_j)$ instead of a sample, a mean-field read-out. *Softmax*: $p(\mathbf{h} = \mathbf{e}_j | \mathbf{v}) = \text{softmax}_j(\mathbf{x})$, exactly one unit active per day, which turns the RBM into a mixture model with one state per day. *ReLU*: rectified Gaussian units [9], $h_j = \min(\max(0, x_j + \epsilon), 5)$, $\epsilon \sim \mathcal{N}(0, 1)$. Crossing visible and hidden combinations gives the ten families of Tab. 2.

C. Training

The likelihood gradient is the difference between a data-driven positive phase and a model-driven negative phase. For NB visibles, the score with respect to η_i is the weighted residual $r_i = \theta_i(v_i - \mu_i)/(\mu_i + \theta_i)$, so

$$\Delta W_{ij} \propto \langle h_j r_i \rangle_+ - \langle h_j r_i \rangle_-, \quad (9)$$

and analogously for $\mathbf{a}$ and $\mathbf{b}$; θ_i (and π_i) are updated by automatic differentiation of the positive-phase NLL with a learning rate ten times smaller. The negative phase uses PCD-1 [7] with 500 persistent chains initialised from training days; at each mini-batch a random subset of chains takes one Gibbs step and is written back.

All families share one protocol (Tab. 1): RMSprop ($\beta = 0.9$) with initial learning rate 0.01 decayed by 0.998 per epoch, 20 mini-batches per epoch with a batch size growing quadratically from 10 to 256, ℓ_1 penalty 10^{-4} on W , and 500 epochs (1000 for ZINB-Sigmoid and ZINB-Softmax). Two guards keep training finite: η_i is clamped at 10 before the exponential and ReLU activations are clamped to $[0, 5]$. Every family is trained with 10 seeds for each L value. In order to keep the training procedure stable and fast, we adopt PCD instead of the basic contrastive divergence, thus making the training almost always complete successfully.

D. Evaluation Protocol and Metrics

In order to determine the number of hidden units L , we compare the performance of the various choices for visible and hidden units representation over the validation set. As for the metric to consider, we utilize the Negative Log-Likelihood (NLL) to perform the comparison, so its minimization produces the best models. We monitor the validation NLL per entry, $-\frac{1}{ND} \sum_{n,i} \log p(v_i^{(n)} | \mathbf{h}^{(n)})$ with $\mathbf{h}^{(n)} \sim p(\mathbf{h} | \mathbf{v}^{(n)})$, a one-step reconstruction likelihood that needs no partition function. It is not normalised: through the $\ln \Gamma(v_i + 1)$ term it depends on the count scale s and on the data, so it has no absolute meaning. We use it only to check convergence and to compare models trained on the same data and scale (choice of L , Tab. 2); the substantive comparisons rest on reconstruction error and on the archetype analysis. Binary models are monitored with the negative pseudo-log-likelihood [10], which is not on the same scale. However, we do not blindly increment L : firstly, since we want to compress the information from the raw data space, it would not make sense to make the number of hidden units greater than the number of visible units (the 83 taxa). Secondly, having a greater L makes the task more computationally intensive and more prone to overfitting. The latter can be spotted when, in the validation set, the NLL starts to plateau or, even worse, to rise. Therefore, when choosing the number of hidden units for each model, we look at the relative improvement with $L + 1$ with respect to L hidden units. This allows us to individuate the values for which the NLL starts to become constant and choose the lower value, since we prefer the simpler model when the performance is comparable and, thus, the improvement is not good enough to justify an increase in complexity.

For Softmax layers a unit is *used* if it is activated at least one time over the considered data set. For Sigmoid read-outs a unit is *on* if $\sigma(x_j) \geq 0.5$, so each day maps to an L -bit pattern.

We compare (i) day labels, the dominant archetype against the Softmax state, by normalised mutual information (NMI, 0 for independent and 1 for identical partitions) and Cluster Purity Standard (CPI, [11]) (share of days whose state's majority archetype is their own); and (ii) profiles, the cosine similarity between each archetype's taxon weights and the visible mean $\mu(\mathbf{e}_j)$ produced by each hidden unit alone. The reference is the archetypal analysis, provided by the research group supervising the course, with $k = 5$ archetypes computed independently on the same series (A1–A5, Section IV-D).

E. Reconstruction protocol

Another task for which we want to utilize our models is reconstructing missing data. In order to test their capability for this assignment, we consider the validation set and for each entry we mask one taxon by setting its abundance to zero, repeating the process for all taxa. Therefore, we evaluate the reconstruction both on the masked species and the non-masked ones via absolute difference with respect to the ground truth (L1 error) and cosine similarity. We report the absolute error on the masked taxon, the mean L1 error and the cosine

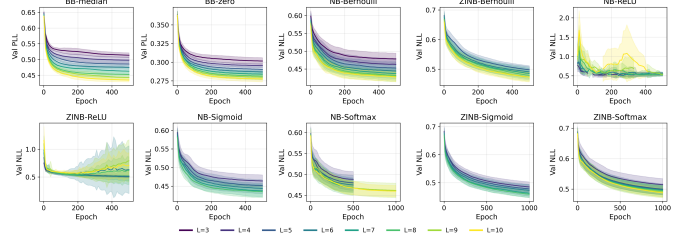


Fig. 2. With PCD, every bounded family trains smoothly up to $L = 10$. Validation metric (NLL for count models, negative PLL for binary ones.) against epoch for each family (panel) and L (colour) for the validation set chosen from shuffled entries of the dataset. Each point is the mean $\pm 1\sigma$ over 10 seeds. Even using clamped ReLU we get instabilities.

similarity on the other taxa, with 20 repetitions per taxon and day, averaged over the 10 seeds. ReLU families (divergent seeds) and binary families (0/1 data) are excluded.

Finally, we also try to perform imputation for those entries for which the abundances are not available, so we do not have a ground truth to compare the reconstructed value to. We consider the 160 days with structurally missing taxa and run our models by initializing the missing values to zero, imputing by Gibbs sampling. The score is the count NLL of the observed taxa under 100 hidden samples, per observed taxon and each family is evaluated at its best L .

IV. RESULTS

A. Training Stability on Raw Counts

The use of PCD allows us to explore greater values for the number of hidden units L (Fig. 2), for which we encounter frequent divergences for $L \geq 8$ when, instead, using CD. We reckon this is, partially, due to the intrinsic multi-modal nature of the abundance of each taxon over time: it presents periodical spikes ("bloom" phases) separated by periods of time for which they are at their baseline abundance ("inter-bloom" phase). Therefore, with PCD we make moving from one mode to the other easier, alleviating the problem of locking the chain to the same mode when sampling. Moreover, as expected due to its unbounded nature, we find that the ReLU activation for hidden units remains unstable; the $[0, 5]$ clamp improves its stability but still produces divergences. Due to this behaviour, we choose to discard it.

B. Choice of L

The results for Negative Binomial (NB) visible units can be seen in Fig. 3. A common (and expected) trend can be seen across the models: increasing the number of hidden units produces an improvement in the NLL. Following the procedure outlined in Section III-D, for NB visible units with Bernoulli or Softmax hidden units, we choose $L = 6$ as the best value, since it is when the NLL starts to plateau. The full comparison for all the models with 6 hidden units can be seen in Tab. 2.

C. How Many Community States

The interpretability of the number of hidden units changes depending on the chosen activation: when we use Bernoulli

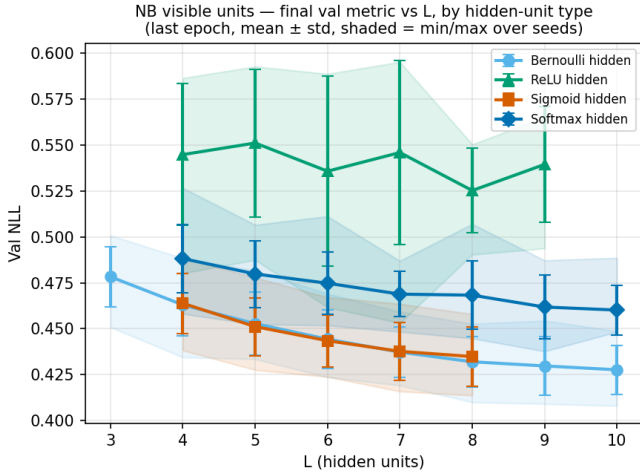


Fig. 3. NB performance on validation set

TABLE 2. All bounded count families fit stably; NB with Bernoulli or Sigmoid hidden performs best. Final validation metric at $L = 6$, shuffled split, mean $\pm$ std over 10 seeds (count models: NLL per entry, comparable only between models on the same scaled data; binary models: negative PLL, a different scale)

Visible	Hidden	Val. metric
NB	Bernoulli	0.444 ± 0.015
NB	Sigmoid	0.443 ± 0.013
NB	Softmax	0.475 ± 0.016
NB	ReLU	0.6 ± 0.1
ZINB	Bernoulli	0.491 ± 0.016
ZINB	Sigmoid	0.473 ± 0.019
ZINB	Softmax	0.498 ± 0.016
ZINB	ReLU	0.6 ± 0.4
Binary (median)	Bernoulli	0.475 ± 0.006
Binary (zero)	Bernoulli	0.287 ± 0.003

each unit is either "on" or "off", resulting in a total of 2^L possible configurations. This means that, given an input, only the switched on hidden units contribute to the output. Instead, when we use a Softmax function for the hidden units, only one unit is assigned to the input (one-hot encoding), so L also becomes the number of states in the latent space. Therefore, we can use the latter as a sort of clustering procedure to individuate in our dataset some fundamental configurations that can explain the vast majority of the whole taxa combinations observed. Thus, using a Softmax hidden layer with $L = 4, \dots, 10$ units, the NB-Softmax RBM follows $y = L$ up to $L \approx 5$ and then saturates (Fig. 4): it uses 5.8 ± 0.8 units at $L = 7$ and 6.2 ± 0.6 at $L = 10$, never fewer than 5 nor more than 7 per seed from $L = 7$ on. The ZINB-Softmax RBM, with a different visible likelihood, gives the same plateau (5.5 ± 0.5 at $L = 7$, 6.5 ± 1.0 at $L = 10$), within one standard deviation of NB at every L . The fact that not all hidden units are utilized for $L \geq 6$ leads us to believe that, under the assumption of a one-to-one relation between archetypes and one-hot encoded hidden units, there exists from

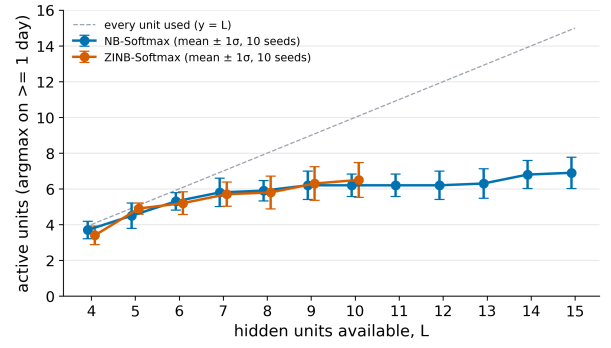


Fig. 4. Offered up to 10 one-hot states, the RBM uses about six, for both NB and ZINB visible units. Number of active Softmax units against L , mean $\pm$ std over 10 seeds, shuffled split.

5 to 7 archetypes in the data.

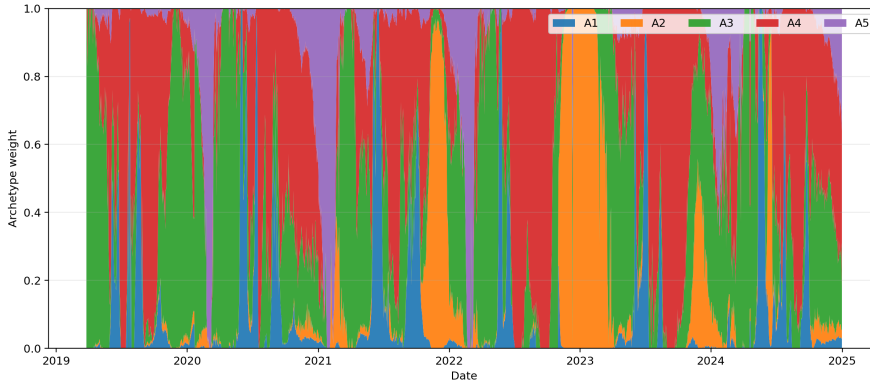
D. Comparison with Archetypes

The reference archetypal analysis, describes each day as a mixture of five archetypes (Fig. 5a). Each is led by one taxon: A1 by *Dinobryon* (80% of its weight), A2 by *Aulacoseira* (96%), A3 by cryptophytes (44%), A4 by colonial green algae with cyanobacteria, A5 by centric diatoms (54%). Their top-5 taxa (Fig. 5b) share a common background: cryptophytes are in all five sets, rotifers and nauplii in three, while the taxa specific to one archetype are what separates them.

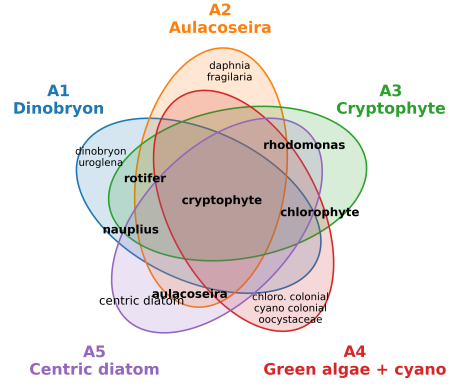
Labelling each day by its dominant archetype and by its Softmax state gives two partitions of the same 1906 days (Fig. 6). Three archetypes are captured by one state each: A2 by h_2 (94% of its days), A5 by h_4 (99%) and A1 by h_5 (96%). However, the states are not exclusive: h_5 also takes 47% of the A4 days and h_4 41% of the A3 days, while A3 and A4 are each split mainly over two states. Overall, NMI is 0.45, which highlights how the archetypes are not completely mapped one-to-one to the hidden units but they do share some similarity in their representations. In fact, when computing the CPI we get a 71% agreement between archetypes and hidden units (so by labeling each hidden unit with its most frequent archetype, the two models predict the same archetype for 71% of the days in the dataset). The figure uses the best seed at $L = 6$, which leaves one unit without days: this is typical at $L = 6$ (5.1 ± 0.7 units used), while the six-state plateau starts at $L = 7$.

When we, instead, consider a Sigmoid activation, a day is a combination of units rather than a single state. All six units are used, with 10 patterns covering 80% of the days and 15 covering 90%, as can be seen in Fig. 7. Therefore, in this case we can not consider a single hidden unit analogous to an archetype, since it is their combinations that produces the output of the RBM. Still, we can observe that out of the possible 64 combinations, only about half are actually used when examining the validation set, which indicates how $L = 6$ is sufficient, as it is with one-hot encoded hidden units, to represent the data.

Nonetheless, we also look for a match between the given archetypes and the patterns utilized by the RBM with the



(a) Archetype weights per day



(b) Top-5 taxa per archetype

Fig. 5. The reference archetypes alternate with the seasons and are each led by one taxon over a shared background of cryptophytes, rotifers and nauplii. (a) Weights of the $k = 5$ archetypes per day (sum to 1). (b) Five-set Venn diagram of the five largest-weight taxa of each archetype; bold labels are taxa shared by two or more archetypes.

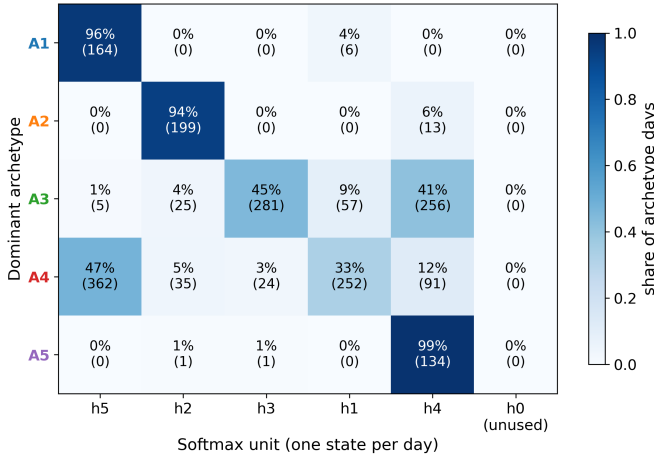


Fig. 6. Softmax states and dominant archetypes mapping is not injective: A1, A2 and A5 each map to one state, A3 and A4 are split and partly merged. NMI 0.45. Units are ordered by the archetype whose days they capture the largest share of and day counts are in brackets.

Sigmoid activation. In order to do this, we compute the cosine similarity between the archetype weights and all the possible $2^L = 64$ combinations of the hidden units to verify if any pattern reconstructs well the archetype. However, this is not sufficient to determine a correlation between them even in the case of high cosine similarity, since it may just be a spurious relation given by the combination of the hidden units. Therefore, to verify such correspondence, we further count the fraction of days for which each of the examined patterns are activated in agreement with their most similar archetype over the entire dataset (Tab. 3). We can see that all five combinations found are patterns that the days actually use (ranked between the second and the eleventh in frequency) and that they give similar results as with Softmax. In fact, A1 and A2 are well matched to a single pattern, A4 only partially (with its most similar pattern covering 44% of its days) while A3 and A4 are spread over many patterns and, therefore, can not be associated with a single one, mirroring what happens

TABLE 3. Every archetype is reproduced by a combination of hidden units, not by a single one. Cosine similarity between each archetype profile and the NB mean generated by the hidden layer set to: no unit on, the best single unit, and the best of the 2^6 combinations. For the best combination: its pattern ($h_0 \dots h_5$), its frequency rank among the 34 patterns the days use (Fig. 7), and the share of the archetype's days whose pattern it is. NB-Sigmoid, $L = 6$, shuffled split, best seed.

	A1	A2	A3	A4	A5
No unit	0.26	0.09	0.17	0.52	0.08
Best unit	0.73	0.54	0.58	0.78	0.62
Best combin.	0.97	1.00	0.99	0.94	0.97
Pattern	000101	101100	011111	010001	100110
Freq. rank	2	3	6	5	11
Days in it	96%	81%	19%	16%	44%

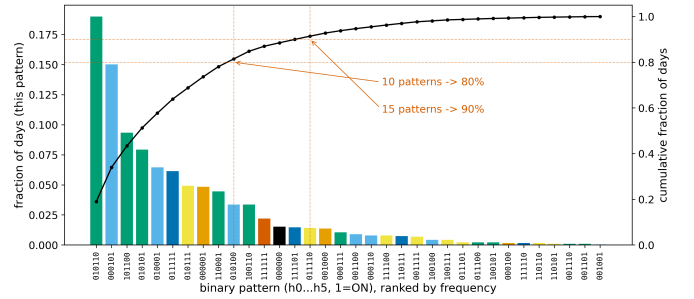


Fig. 7. NB-Sigmoid $L = 6$, best seed, shuffled split. A day is a combination of units: 34 of 64 patterns occur and 10 cover 80% of the days. Frequency of each 6-bit pattern, a unit being on if $p(h_j=1 | \mathbf{v}) \geq 0.5$ (bars, coloured by number of active units), and cumulative coverage (line)

for the same archetypes in the one-hot encoding case.

E. Reconstruction and Imputation

We measure the performance of our models in reconstructing missing data as described in Section III-E. The results are summarized in Tab. 4: NB with Sigmoid is the first or tied on every metric. Moreover, we can see how, for this task, using Softmax consistently produces worse results (both on the masked and non-masked species) than Sigmoid or Bernoulli, highlighting how one-hot encoding is not the best solution for reconstruction of missing data. Finally, across all models,

TABLE 4. Masked-taxon reconstruction: NB-Sigmoid is first or tied; the one-hot Softmax layer costs about 30% on the unmasked taxa. Mean absolute error (scaled counts) and cosine similarity, $L = 6$, mean $\pm$ std over 10 seeds, each scored on its own validation days.

Family	Masked L1	Other L1	Cosine
NB-Bernoulli	0.609 ± 0.022	0.516 ± 0.012	0.860 ± 0.010
NB-Sigmoid	0.604 ± 0.022	0.509 ± 0.017	0.869 ± 0.008
NB-Softmax	0.700 ± 0.052	0.679 ± 0.055	0.761 ± 0.042
ZINB-Bernoulli	0.613 ± 0.023	0.534 ± 0.025	0.846 ± 0.014
ZINB-Sigmoid	0.604 ± 0.027	0.514 ± 0.027	0.869 ± 0.021
ZINB-Softmax	0.683 ± 0.031	0.667 ± 0.037	0.760 ± 0.022

the taxa that are the hardest to reconstruct when masked are *Aulacoseira*, *Dinobryon* and *cryptophytes*. One such example can be seen in Fig. 8, where we show the reconstruction for a randomly selected validation sample for Bernoulli and Sigmoid hidden units (for which we expect similar results). Most taxa are reconstructed within a fraction of a count, but *Aulacoseira* is overestimated, about fivefold for both visible layers. This single taxon accounts for most of the day’s mean L1 error: 1.83 and 1.99 over the 83 taxa, 0.19 and 0.17 without it.

We perform the imputation task on the 160 days with missing taxa. The results, in terms of log-likelihood, can be seen in Fig. 9, grouped by the number of missing taxa in the input. All models obtain similar results, with NB-Sigmoid usually getting the best mean value, but still well within 1 standard deviation of the others, so there is no clear standout winner.

F. Failure Modes: Zero Inflation

The ZINB visible units are designed to model the zero-inflation of the data, however, in our work it does not outperform the Negative Binomial for visible units: at 500 epochs its NLL, θ and π are all still moving and close to the NB values, indicating that it may not have yet reached convergence. While this can be trivially solved by increasing the training epochs, the model still requires more fine-tuning in order to possibly beat the NB mode, which we consider working the best. More specifically, π and θ require careful tuning, making the interpretation of the results of the RBM more difficult to understand as they vary. Nonetheless, it is still an interesting approach which may be expanded upon in future works to outperform NB.

V. CONCLUSIONS

We trained restricted Boltzmann machines with binary, Negative Binomial and Zero-Inflated NB visible units and four hidden-unit types on almost six years of daily plankton counts from Lake Greifen, made training on raw counts stable, and compared the learnt latent states with an archetypal analysis computed independently on the same data.

Moreover, we found that count-likelihood RBMs can recover plankton community states that the independent method also finds: while there is not an exact one-to-one relation between archetypes and hidden units, they do share some common information. In fact, a one-hot hidden layer settled

on about 6 or 7 states whatever the visible likelihood, and its day labels matched the dominant archetype (NMI 0.45, purity 71% for $L = 6$ but only 5 units were actually active). Similarly, the final NB-Sigmoid model, with 6 hidden units, reproduced every archetype with a combination of units, lacking a specific pattern of hidden units activated only for A3 and A4, for which the RBM associated multiple hidden unit configurations, as the Softmax states did.

As for the choice of the hidden layer, we noticed a measurable trade-off: while one-hot states were easier to interpret as different community states, they costed about 30% when doing reconstruction on masked taxa with respect to Sigmoid/Bernoulli hidden units, with Sigmoid being preferable due to its smoother results. Moreover, we found that doing multiple PCD steps would progressively drift the reconstructed taxon abundance further away from the ground truth, so we limited our analysis to PCD-1.

Meanwhile, a limitation we encountered when doing imputation on completely missing values and initializing them at zero was that there was no ground truth to compare them to, so their scores and ranking are just relative. We also opted, as we did for the reconstruction, to use PCD-1, so a different number of iterations may affect the results.

Finally, our NB-Softmax analysis shows a saturation in the utilization of hidden units at 6 for $L \geq 8$, highlighting how the underlying structure of the data may be organized into 6 fundamental states, which is comparable to the independent 5 archetype study.

Future works on this topic may tackle the fine-tuning and training of ZINB to convergence, with longer runs, better mixing and optionally introducing a hidden-dependent π to better deal with the zero inflated counts. Alternatively, another interesting method would be to completely change the architecture and approach the problem with a variational autoencoder, tuning the latent space dimension to see if it replicates the six-state plateau we found with NB-Softmax.

DATA AND CODE AVAILABILITY

The plankton time series is the Lake Greifen imaging dataset described in [1]. Code, configurations and every script that produced the figures and tables are available on GitHub; trained weights are not distributed and are regenerated by the training script.

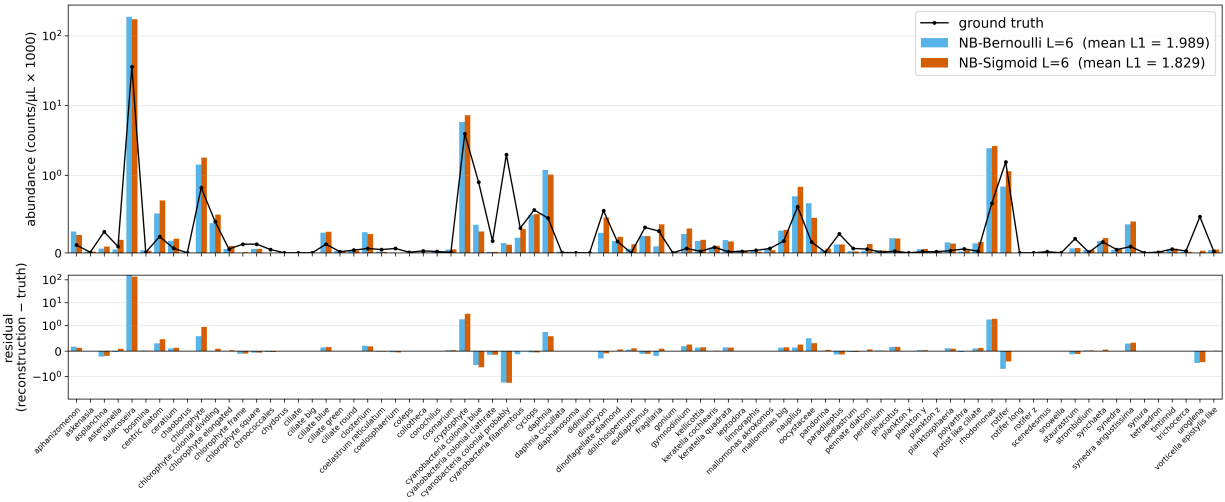


Fig. 8. On a random validation day most taxa are reconstructed closely and the error comes from *Aulacoseira*, overestimated about fivefold. Ground truth (line) and one PCD step reconstruction (bars) by NB RBMs with $L = 6$. Abundances are in log-scale.

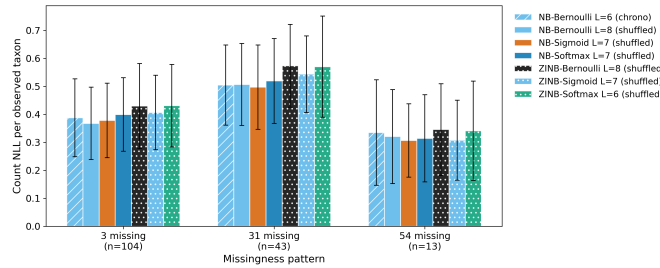


Fig. 9. Imputation on days with missing taxa: differences are smaller than the day-to-day spread. Count NLL per observed taxon after 1 PCD step, by missingness pattern; each family at its best L ; bars show mean $\pm$ std over days. Hatching marks the chronological split (//) and ZINB(dots).

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